UNIT - 1
Introduction

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Reference Books

1. **Data Science for Business** – Foster Provost and Tom Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
2. **Python for Data Analysis** – Wes McKinney, *Python for Data Analysis*, 2nd Edition, O'Reilly Media, 2017.
3. **Doing Data Science** – Cathy O'Neil and Rachel Schutt, *Doing Data Science*, O'Reilly Media, 2013.
4. **Data Science from Scratch** – Joel Grus, *Data Science from Scratch*, 2nd Edition, O'Reilly Media, 2019.
5. **Big Data Fundamentals** – Thomas Erl, Wajid Khattak, and Paul Buhler, *Big Data Fundamentals*, Prentice Hall, 2015.

Data Science

Data

Raw, fact, figures, symbols, unorganized or **unstructured** fact is called data.

Or we can say - Data are individual units of information. it can be generated from different sources like **sensors**, **surveys**, **transactions**, **social media** etc.

- Data contains **numbers, text, observation, statements** and **characters** in a raw form.
- Data is meaningless.
- Data does not directly help in decision making.

Types of Data

Quantitative data	Numerical form, like 50 Kg, 165 cm, 15887 etc.
Discrete Data	Whole numbers. example the number of employees in a department.
Continuous Data	Data that can take any value within a range. For example, wind speed, and temperature.
Qualitative data	It's available in a descriptive form for example name, gender, address, and features of a person.
Nominal Data	Data that represents categories with no inherent order. For example, colours, and gender.

Types of Data

Ordinal Data	Data that represents categories with a specific order or ranking. For example, ranking satisfaction levels as "poor," "average," or "excellent."
Categorical Data	The data which represents categories or labels and is often qualitative is called categorical data. It can include nominal and ordinal data.
Numerical Data	This type of data includes numbers.
Time Series Data	Data collected over time intervals like stock prices, weather data, and sales figures.
Spatial Data	Data associated with geographic locations like Google maps, GPS data, and satellite images.

Information

Processed data is called information. Information is presented in the form of **words, language, thoughts** and **ideas**.

- Information is organized.
- It is meaningful.
- It is always useful and used in decision-making.
- Information depends on Data.
- Example of information: Merit List, receipts, reports, report cards etc.

Science

Science is the **systematic study of the natural world** through **observation, experimentation, analysis** and **logical reasoning**. It aims to understand how things work and to discover facts, principles and laws that explain natural phenomena.

Data Science

Data Science is an **interdisciplinary** field that **combines** **statistics**, **mathematics**, **programming**, **machine learning** and **domain knowledge** to **collect**, **process**, **analyze** and **interpret data**. Its main **purpose** is to extract useful information and insights from large amounts of data and use them for **decision-making**.

Data Science

Data Science is the **study of data** using **scientific methods**, **algorithms** and **computational techniques** to discover **valuable insights**, **solve problems** and **make predictions**.

It involves:

- ✓ Collecting data
- ✓ Cleaning data
- ✓ Analyzing data
- ✓ Visualizing results
- ✓ Building predictive models
- ✓ Making data-driven decision

Real-Life Example of Data Science

Netflix Recommendation System

Netflix uses Data Science to recommend movies and TV shows.

How It Works:

Step 1: Collect Data

Netflix collects:

- Viewing history
- Search history
- Ratings
- Watch time

Real-Life Example of Data Science

Step 2: Analyze Data

The system analyzes user preferences.

Step 3: Build Recommendation Model

Machine Learning algorithms identify similar viewing patterns.

Step 4: Suggest Content

Netflix recommends movies based on user interests.

Result:

Users receive personalized recommendations.

Application of Data Science

1. Healthcare

- Disease prediction
- Medical diagnosis
- Patient monitoring

Example:

Predicting the risk of heart disease.

Application of Data Science

2. Banking

- Fraud detection
- Credit scoring
- Risk analysis

Example:

Detecting suspicious transactions.

Application of Data Science

3. E-Commerce

- Product recommendations
- Customer segmentation
- Demand forecasting

Example:

Amazon recommending products.

Application of Data Science

4. Education

- Student performance analysis
- Learning recommendations

Example:

Identifying students who need additional support.

Application of Data Science

5. Social Media

- Sentiment analysis
- Content recommendations

Example:

Facebook suggesting friends.

Why is Data Science Important?

Data Science helps organizations-

1. Make better decisions
2. Predict future trends
3. Improve business performance
4. Understand customer behavior
5. Detect fraud and risks
6. Automate processes using Artificial Intelligence (AI)

Components of Data Science

Data Science combines several disciplines:

1. **Mathematics**: Provides formulas and models for analysis.
2. **Statistics**: Helps understand patterns and trends.
3. **Programming**: Used to process and analyze data.

   Example- Popular languages: **Python**, **R**
4. **Machine Learning**

Enables systems to learn from data and make predictions.

5. **Domain Knowledge**

Understanding the business or industry where data is applied.

Key Feature of Data Science

- Uses data to solve problems
- Combines statistics and programming
- Supports machine learning and AI
- Helps in prediction and decision-making
- Works with large datasets

Tools Used in Data Science

Tool	Purpose
Python	Programming and Machine Learning
R	Statistical Analysis
SQL	Database Management
Excel	Data Analysis
Tableau	Data Visualization
Power BI	Dashboard Creation

Data Science Process

The Data Science lifecycle consists of several stages:

Data Collection
↓
Data Cleaning
↓
Data Analysis
↓
Data Visualization
↓
Model Building
↓
Decision Making

Data Science Process



Problem Definition
Define the question or goal



Data Collection
Gather relevant information



Data Cleaning
Prepare data by fixing errors



Exploratory Data Analysis
Understand patterns in data



Modeling
Build predictive models



Evaluation
Test model performance



Deployment
Put model into use



Insights/Action
Use findings to make decisions

Data Science Process

1. Data Collection

The first step is gathering data from different sources.

Sources of Data:

1. Databases
2. Websites
3. Social media
4. Mobile apps
5. Sensors
6. Excel files

Example:

An online shopping company collects customer purchase records.

Data Science Process

2. Data Cleaning

Raw data often contains:

1. Missing values
2. Duplicate records
3. Incorrect entries

Data cleaning improves data quality before analysis.

Example:

Removing duplicate customer records from a database.

Data Science Process

3. Data Analysis

In this stage, data is examined to identify patterns and relationships.

Techniques Used:

1. Statistical analysis
2. Trend analysis
3. Correlation analysis

Example:

Finding which products are most frequently purchased together.

Data Science Process

4. Data Visualization

Data is presented visually using graphs and charts to make it easier to understand.

Common Visualization Tools:

1. Bar Charts
2. Histograms
3. Scatter Plots
4. Heat Maps
5. Box Plots

Example:

Displaying monthly sales using a bar chart.

Data Science Process

5. Model Building

Machine Learning algorithms are used to build predictive models.

Examples:

- Customer churn prediction
- Sales forecasting
- Fraud detection

6. Decision Making

The final insights help businesses make informed decisions.

Example:

Increasing inventory for products expected to have higher demand.

Types of Data

Data can be broadly categorized into three types **based on its organization** and **structure**.

1. **Structured Data**
2. **Semi-structured Data**
3. **Unstructured Data**

Types of Data

1. Structured Data

Structured data is **highly organized** and **follows a predefined schema** or **format**. It is typically **stored in relational databases** or **tabular formats**, where **data is organized into rows and columns**.

Example:

1. **Databases**: Tables in relational databases (e.g., customer data, product catalogs).
2. **Spreadsheets**: Excel or Google Sheets files with rows and columns.
3. **CSV Files**: Comma-separated values files with a clear structure.

Types of Data

Characteristics of Structured Data :

1. **Fixed Schema**: Data adheres to a strict schema with defined data types, relationships, and constraints.
2. **Tabular Format**: Organized into rows and columns, making it easy to query and analyze.
3. **Stored in Relational Databases**: Commonly stored in SQL-based databases like MySQL, PostgreSQL, or SQL Server.
4. **Easy to Process**: Can be easily processed using SQL or other structured query tools.

Types of Data

2. Semi Structured Data

Semi-structured **data does not follow a strict schema** like structured data **but still has some level of organization**. It often contains **tags, markers**, or **metadata** that help define its structure.

Example:

JSON Files: **J**ava**S**cript **O**bject **N**otation files with key-value pairs.

```
{    "name": "Raj",
     "age": 30,
     "address": {
   "city": "Chennai",
   "pincode": "600024"
 }  }
```

Types of Data

2. Semi Structured Data

Example:

XML Files: E**x**tensible **M**arkup **L**anguage files with nested tags.

```
<person>
  <name>Raj</name>
  <age>30</age>
  <address>
    <city>Chennai</city>
    <pincode>600024</pincode>
  </address>
</person>
```

Log Files: Server or application logs with varying fields.

Types of Data

Characteristics of Semi Structured Data :

1. **Flexible Schema**: No fixed schema, but it may have self-describing elements like tags or keys.
2. **Hierarchical or Nested Structure**: Often represented in formats like JSON, XML, or YAML.
3. **Stored in NoSQL Databases**: Commonly stored in NoSQL databases like MongoDB, Cassandra, or Elasticsearch.
4. **Easier to Scale**: More flexible than structured data, making it easier to scale and adapt to changes.

Types of Data

3. Unstructured Data

Unstructured data has **no predefined structure** or **schema**. It is often **raw** and **unorganized**, making it the most challenging type of data to process and analyze.

Example:

- **Text Data**: Emails, social media posts, word documents.
- **Multimedia**: Images, videos, audio files.
- **Sensor Data**: Raw data from IoT devices without metadata.

Types of Data

Characteristics of Unstructured Data :

1. **No Schema**: No fixed format or organization.
2. **Diverse Formats**: Can include text, images, videos, audio, and more.
3. **Stored in Data Lakes or File Systems**: Often stored in data lakes (e.g., AWS S3, Azure Data Lake) or file systems.
4. **Requires Advanced Processing**: Needs tools like natural language processing (NLP), computer vision, or machine learning to extract insights.

Comparison of Data Types

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Schema	Fixed and predefined	Flexible, self-describing	No schema
Format	Tabular (rows/columns)	JSON, XML, YAML	Text, images, videos
Storage	Relational databases	NoSQL databases	Data lakes, file systems
Querying	Easy (SQL)	Moderate (requires parsing)	Difficult (requires advanced tools)
Examples	SQL tables, CSV files	JSON, XML, log files	Emails, images, videos
Use Cases	Financial records, CRM	Web APIs, IoT data	Social media, multimedia analysis

Advantages and Disadvantages of Data Science

Advantages

1. **Better Decision Making**: Data-driven decisions
2. **Accurate Predictions**: Forecast future events
3. **Versatile Applications**: Health, Finance, Media
4. **Customer Insights**: Understands user behavior
5. **Automation of Tasks**: increases efficiency and reduces human errors
6. **High Demand and Career Opportunities**
7. **Scope for Innovation**

Advantages and Disadvantages of Data Science

Disadvantages

1. **High Cost**: Requires advanced tools and infrastructure
2. **Data Privacy Issues**: Sensitive information may be exposed
3. **Complex Implementation**: Requires skilled professionals
4. **Data Quality Dependency**: Poor data leads to poor results
5. **Dependence on Data:** inaccurate or incomplete
6. **Job Saturation at the Entry Level:** becoming competitive
7. **Not Always Accurate**: predictions are not always 100% accurate

Digital Universe

The Digital Universe refers to the **huge amount of digital data** that is **created, collected, stored** and **shared** worldwide every day. This data is generated from various **sources** such as **smartphones, social media platforms, websites, sensors, online transactions, emails, videos** and **Internet of Things (IoT) devices.**

In Data Science, the Digital Universe serves as the primary source of data that is analyzed to extract meaningful insights, identify patterns, and support decision-making.

Digital Universe

The Digital Universe is the **collection of all digital data** produced by **people, organizations** and **machines**.

Sources of the Digital Universe

Digital data is generated from numerous sources, including:

1. **Social Media:** Facebook, Instagram,Twitter (X), LinkedIn
2. **E-Commerce Platforms:** Amazon, Flipkart, eBay
3. **Mobile Devices:** Smartphones, Tablets
4. **Internet of Things (IoT):** Smart watches, Smart home devices, Sensors
5. **Business Transactions:** Banking records, Online payments, Customer databases
6. **Multimedia Content:** Photos, Videos, Audio files

Evolution of the Digital Universe

The growth of the Digital Universe has accelerated due to technological advancements such as:

1. **Internet expansion**
2. **Smartphones**
3. **Cloud computing**
4. **Social media platforms**
5. **Internet of Things (IoT)**
6. **Artificial Intelligence (AI)**

Today, data is generated at a much faster rate than ever before.

Characteristics of the Digital Universe

1. Massive Volume

The Digital Universe contains an enormous amount of data generated from multiple sources worldwide.

Example: Social media platforms generate terabytes of data every day.

2. Rapid Growth

The amount of digital data is increasing continuously due to internet usage and connected devices.

Example: Every new smartphone user contributes additional data through browsing, messaging, and app usage.

Characteristics of the Digital Universe

3. Variety of Data

Data exists in multiple formats:

- ✓ Structured data
- ✓ Semi-structured data
- ✓ Unstructured data

Example: A bank stores customer details in tables (structured), while customer emails are unstructured.

Characteristics of the Digital Universe

4. Continuous Generation

Digital data is produced every second.

Example: Online purchases, GPS tracking and social media activities generate real-time data.

Digital Universe and Big Data

The Digital Universe is closely connected to **Big Data** because the data generated is:

1. **Volume**: Huge **quantities** of data.
2. **Velocity**: Generated at **high speed**.
3. **Variety**: Available in **different formats**.
4. **Veracity**: Data **quality** and **reliability**.
5. **Value**: **Useful insights** obtained from data.

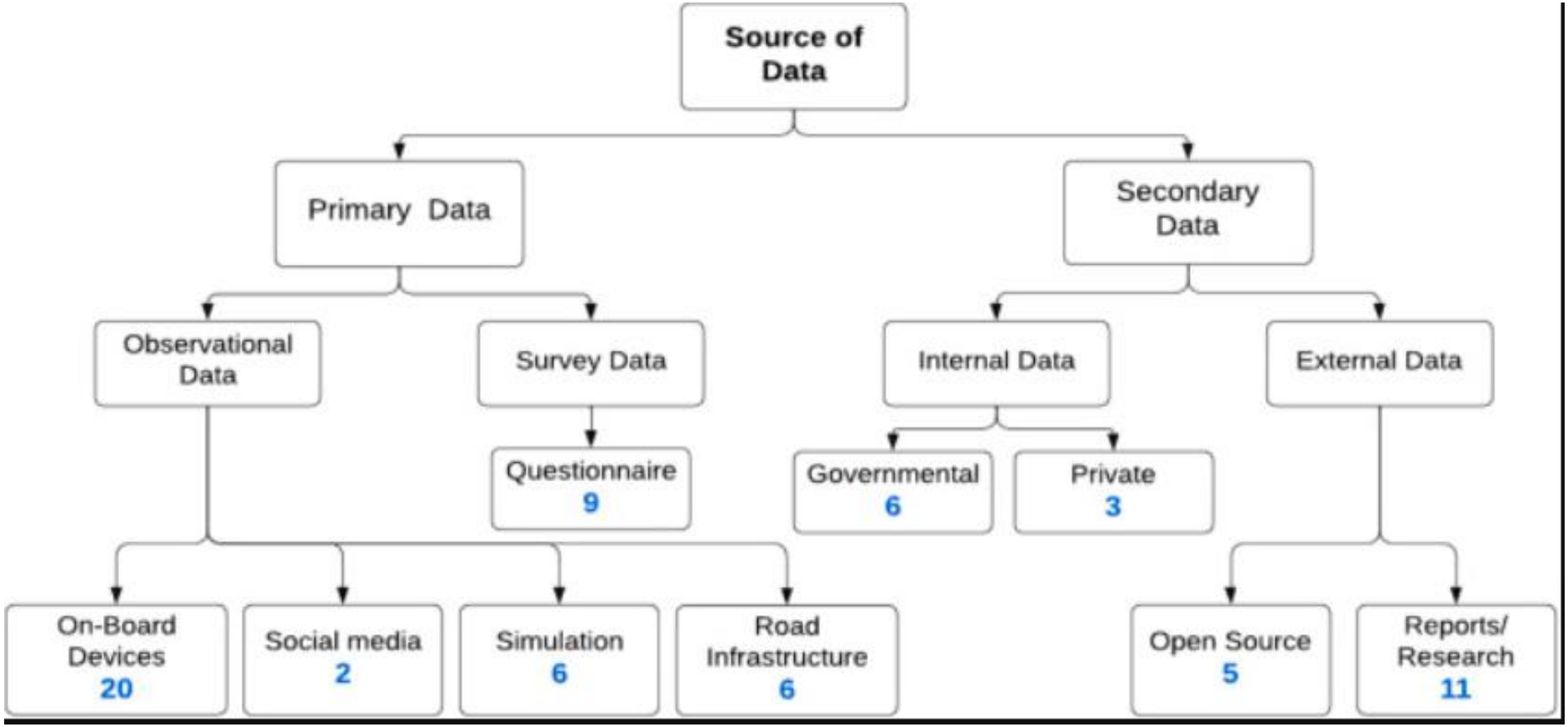
Advantages of Digital Universe

Advantage	Description
Better Decision Making	Provides data-driven insights
Improved Customer Experience	Enables personalized services
Supports AI and Machine Learning	Provides large datasets for training
Business Growth	Helps identify opportunities
Real-Time Analysis	Supports instant decision-making

Challenges of Digital Universe

Challenge	Description
Data Storage	Huge storage requirements
Data Security	Risk of cyber attacks
Data Privacy	Protection of personal information
Data Quality	Incomplete or inaccurate data
Data Management	Difficult to organize massive datasets

Sources of Data



Sources of Data

In Data Science, data can be collected from various **internal** and **external sources**. These sources provide valuable information that helps organizations understand trends, customer behavior, business performance and future opportunities.

The actual data is then further divided mainly into two types known as:

1. **Primary data**
2. **Secondary data**

Sources of Data

1. Primary data

The data which is **Raw, original** and **extracted** directly from the **official sources** is known as primary data.

This type of data is collected directly by performing techniques such as:

- ✓ Questionnaires
- ✓ Interviews
- ✓ Surveys
- ✓ Observation
- ✓ Experimental

Sources of Data

2. Secondary data

Secondary data is the data which has **already** been **collected** and **reused** again for **some valid purpose**. This type of data is previously recorded from primary data.

It has two types:

i. **Internal source**

ii. **External source**

Sources of Data

2. Secondary data

i. Internal source

These types of data can easily be **found within the organization** such as **market record, a sales record, transactions, customer data, accounting resources** etc.

The cost and time consumption is less in obtaining internal sources.

These **sources provide information** about the **organization's operations, customers** and **performance.**

Sources of Data

2. Secondary data

i. Internal source

Example:

A. Sales Data

Sales records contain information about products, customers and transactions.

Example:

- Product purchased
- Quantity sold
- Sales amount

Sources of Data

2. Secondary data

i. Internal source

Example:

B. Customer Databases

Organizations maintain customer information in databases.

Example:

- Customer Name
- Address
- Contact Number
- Purchase History

Sources of Data

2. Secondary data

i. Internal source

Example:

C. Financial Records

Financial departments generate data related to:

- Income
- Expenses
- Profit
- Investments

Sources of Data

2. Secondary data

i. Internal source

Example:

D. Human Resource (HR) Data

HR systems maintain employee information.

Example:

- Employee ID
- Salary
- Attendance
- Performance Records

Sources of Data

2. Secondary data

ii. External source

The data which can't be found at internal organizations and **can be gained through external third party resources** is external source data. The cost and time consumption is more because this contains a huge amount of data.

Example: Government publications, news publications, Registrar General of India, planning commission, international labor bureau, syndicate services and other non-governmental publications.

Sources of Data

Other source

Example: Sensors data, Web traffic, Satellites data etc.

Information Commons

An Information Commons is a **shared environment** where **people can access**, **organize, analyze** and **use information resources, technology** and **services from a single platform**.

In Data Science, Information Commons provides access to data, computing resources, software tools and collaborative spaces that help users transform raw data into meaningful insights.

Information Commons

Information Commons is a **shared space** where **users can access data, information** and **analytical tools for research** and **problem-solving**.

Example-

A group of students analyzing climate change data may use:

1. Government weather datasets
2. Python for analysis
3. Tableau for visualization

The Information Commons provides all these resources in one place.

Information Commons

Example- University Research Center

A university establishes an Information Commons where students and researchers can:

1. Access research datasets
2. Use statistical software
3. Perform machine learning experiments
4. Collaborate on projects

Features of Information Commons

1. **Centralized Access**

   Provides a single location to access information and tools.

2. **Collaboration**

   Encourages teamwork among researchers and analysts.

3. **Resource Sharing**

   Allows multiple users to utilize common resources.

4. **Technology Integration**

   Combines data, software and computing facilities.

5. **Knowledge Creation**

   Supports research and innovation.

Components of Information Commons

1. Data Resources
2. Technology Infrastructure
3. Data Analysis Tools
4. Information Services
5. Collaborative Workspace

Components of Information Commons

1. Data Resources

These include various datasets collected from different sources.

Examples:

- Government databases
- Research datasets
- Business records
- Social media data

Components of Information Commons

2. Technology Infrastructure

Provides the hardware and software required for data analysis.

Examples:

- Computers
- Servers
- Cloud platforms
- High-performance computing systems

Components of Information Commons

3. Data Analysis Tools

Software used for processing and analyzing data.

Examples:

- Python
- R
- Tableau
- Power BI
- SQL

Components of Information Commons

4. Information Services

Support services provided by experts.

Examples:

- Data consultation
- Research assistance
- Technical support

Components of Information Commons

5. Collaborative Workspace

Allows users to work together on data-driven projects.

Examples:

- Shared laboratories
- Digital collaboration platforms
- Research centers

Challenges of Information Commons

1. Data privacy and security concerns
2. Managing large volumes of data
3. Maintaining technology infrastructure
4. Ensuring data quality
5. Access control and authorization

Applications of Information Commons

1. Education

- ✓ Research projects
- ✓ Data science learning
- ✓ Academic collaboration

2. Business

- ➢ Market analysis
- ➢ Customer behavior studies
- ➢ Decision support systems

Applications of Information Commons

3. Healthcare

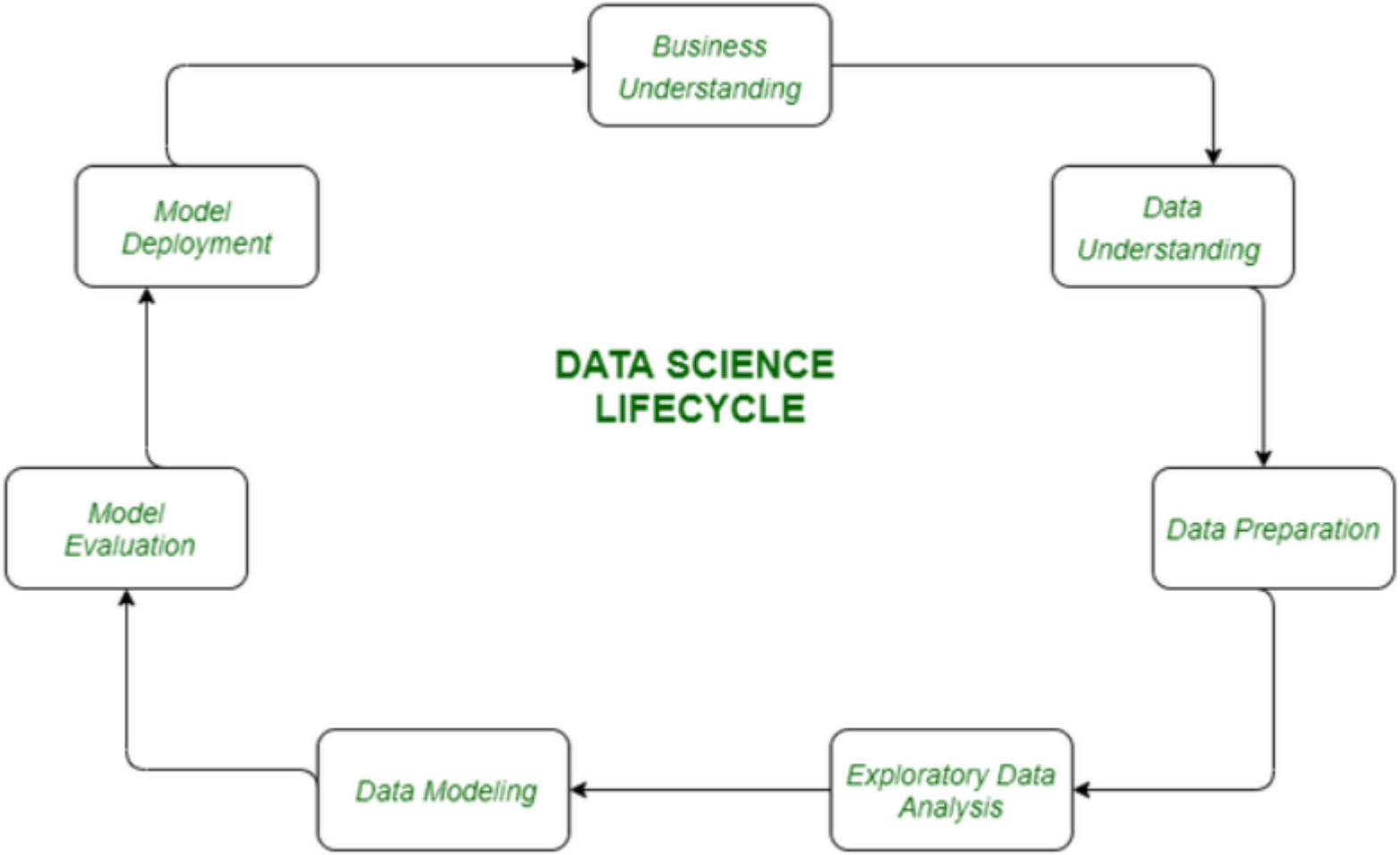
- Medical research
- Disease prediction
- Patient data analysis

4. Government

- Public policy analysis
- Population studies
- Economic planning

Data Science Project Life Cycle

The Data Science Project Life Cycle **(DSPLC)** is a **structured process** followed by **data scientists to solve real-world problems using data**. It consists of a series of stages that **transform raw data** into **valuable insights** and **actionable solutions**.


Business Understanding
Data Understanding
Data Preparation
Exploratory Data Analysis
Data Modeling
Model Evaluation
Model Deployment
DATA SCIENCE LIFECYCLE

Data Science Project Life Cycle

1. Business Understanding
2. Data Preparation (Data Cleaning)
3. Data Collection
4. Exploratory Data Analysis (EDA)
5. Feature Engineering
6. Model Building
7. Model Evaluation
8. Deployment

Data Science Project Life Cycle

1. Business Understanding:

The first step is to understand the business problem and project objectives. Or A clearly defined problem statement.

1. Identify the problem
2. Define project goals
3. Understand business requirements
4. Determine success criteria

Example:

An e-commerce company wants to predict which customers are likely to stop purchasing products.

Data Science Project Life Cycle

2. Data Preparation (Data Cleaning)

Prepare data for analysis by removing errors and inconsistencies.

Activities:

- Handle missing values
- Remove duplicate records
- Correct inaccurate data
- Standardize formats

Data Science Project Life Cycle

3. Data Collection

Gather relevant data from various sources.

Sources of Data: Databases, Websites, Social media, Sensors, Excel files etc.

Example:

Collecting customer purchase history from an online shopping platform.

Data Science Project Life Cycle

3. Data Collection

Example:

Before Cleaning:

Name	Age
Rahul	25
Rahul	25
Priya	Null

After Cleaning:

Name	Age
Rahul	25
Priya	22

Data Science Project Life Cycle

4. Exploratory Data Analysis (EDA)

Understand data characteristics and identify patterns.

Activities:

1. Statistical analysis
2. Data visualization
3. Correlation analysis
4. Trend identification

Tools: Python, Pandas, Matplotlib, Seaborn

Example:

Analyzing customer spending behavior.

Data Science Project Life Cycle

5. Feature Engineering

Create or select important variables that improve model performance.

Activities:

- Feature selection
- Feature transformation
- Creating new features

Example:

Original Features: **Date of Birth**

New Feature: **Age**

Data Science Project Life Cycle

6. Model Building

Develop machine learning models to solve the problem.

Common Algorithms:

Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Means Clustering algorithms

Example:

Building a customer churn prediction model.

Data Science Project Life Cycle

7. Model Evaluation

Assess the performance of the developed model.

Evaluation Metrics:

Accuracy, Precision, Recall, F1-Score

Example:

A model achieves 92% accuracy in predicting customer churn.

Data Science Project Life Cycle

8. Deployment

Make the model available for real-world use.

Methods:

- Web applications
- Mobile applications
- Cloud platforms
- APIs

Example:

Deploying a recommendation system on an e-commerce website.

OSEMN Framework

O – **O**btain (Data Collection)

S – **S**crub (Data Cleaning and Preparation)

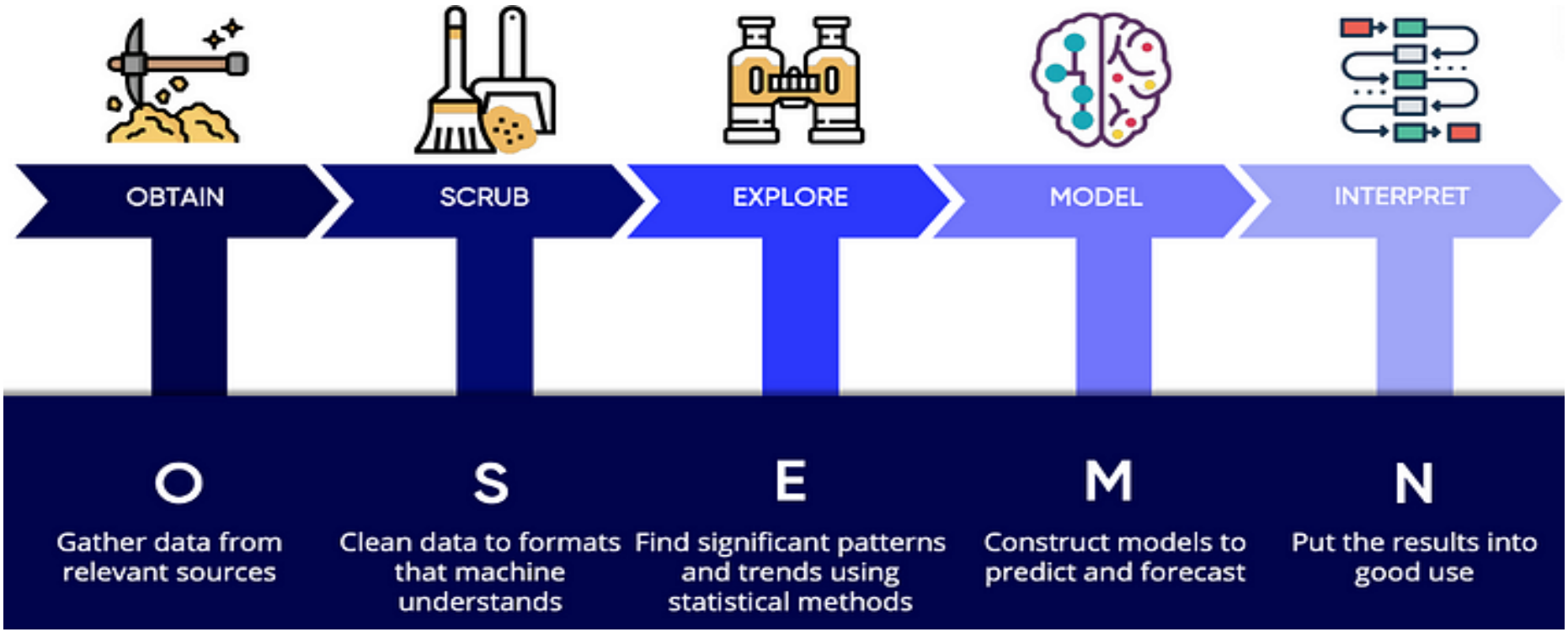
E – **E**xplore data (Exploratory Data Analysis - EDA)

M – **M**odel the data (Machine Learning and Prediction)

N – i**N**terpret and communicate results (Interpretation and Communication)



OSEMN Framework



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